

RUSHI ASHOK SAVANI

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SUMMARY

AI/ML Engineer with experience building machine learning models, LLM applications, retrieval workflows, and Dockerized backend services for operational decision support. Worked across model evaluation, SQL-backed AI systems, cloud deployment, and stakeholder-facing analytics, with a focus on reliable outputs for business and technical users.

EDUCATION

Syracuse University, NY, USA

Masters | Information Systems, Data Science

Aug 2024 - May 2026

(GPA: 3.85/4)

Thadomal Shahani Engineering College, Mumbai, India

Bachelors | Information Technology (*Minors - Artificial Intelligence and Data Science*)

Aug 2020 - Jun 2024

(GPA: 8.52/10)

WORK EXPERIENCE

AI/ML Engineer Co-op

ProHealth Group

Jun 2025 – May 2026

Atlanta GA

- Designed an end-to-end LLM assistant on a LangGraph multi-step architecture, using hybrid retrieval over schema metadata and business-rule context before SQL generation, improving validated answer accuracy from 68% to 92%.
- Built an LLM anomaly investigation pipeline over PostgreSQL operational and financial queries, generating cited root-cause reports that surfaced ~\$70K in preventable annual loss.
- Evaluated LLM outputs with DeepEval test sets, human-in-the-loop review, and prompt versioning to reduce hallucinations and improve response reliability.
- Improved rehospitalization model recall from 62% to 85% with class-weighted XGBoost training, temporal validation splits, and correlation-based feature pruning.
- Tuned decision thresholds with cost functions and isotonic regression calibration to balance recall, precision, and probability scoring across patient-risk bands.
- Collaborated on deployment of ML and LLM workflows as Dockerized FastAPI services on AWS/EKS, supporting GitHub Actions CI/CD, rate limiting, autoscaling, and MLflow experiment tracking.

Data Scientist - Research Assistant

Syracuse University

Jan 2025 – May 2025

Syracuse, NY

- Processed and labeled 5,000+ survey responses through an NLP pipeline, then fine-tuned RoBERTa with weighted cross-entropy, AdamW, and early stopping, reaching 91% cross-validated accuracy.
- Validated classifier performance with stratified K-fold splits and confusion-matrix review, then used Sentence-BERT and HDBSCAN clustering to surface AI-adoption themes in Streamlit dashboards.

Data Analyst

Krishna Jewels

Jun 2023 - Jun 2024

Mumbai, IN

- Converted paper-based inventory, sales, and supplier records into a structured MySQL data layer, migrating 40K+ records and cutting lookup time by 70% through indexing and query tuning.
- Implemented Power BI analytical models with DAX measures for sales, inventory, margin, and profitability KPIs, creating reusable metric logic across dashboards.
- Partnered with stakeholders to define dashboard requirements, automating SQL ETL pipelines and Power BI dashboards that saved 15+ hours weekly and reduced maintenance to one owner.

TECHNICAL SKILLS

- Programming & Backend:** Python, SQL, TypeScript, REST APIs, PostgreSQL, MySQL, Git
- Machine Learning:** Scikit-learn, XGBoost, PyTorch, Hugging Face Transformers, RoBERTa, Sentence-BERT, Model Evaluation, Feature Engineering, MLflow
- LLM & Retrieval Systems:** LangChain, LangGraph, Retrieval-Augmented Generation (RAG), Prompt Engineering, Vector Search, Vector Database, Vertex AI, OpenAI
- Cloud & MLOps:** AWS, Docker, Kubernetes/EKS, GitHub Actions CI/CD, FastAPI Deployment
- Data Engineering & BI:** Data Modeling, ETL/ELT Pipelines, Power BI (DAX), Streamlit

PROJECTS

PrivateRAG | Fully Local Private-Document Q&A System, [Github](#)

- Engineered a fully local RAG system for private-document Q&A, combining document ingestion, chunking, Qdrant vector indexing, metadata filtering, and Llama 3.1 inference without external LLM APIs.
- Delivered a TypeScript chat interface over a local OpenAI-compatible backend, enabling streaming responses, metadata-filtered retrieval, source attribution, and chunk-level traceability.

KarmaScore Predictor | Subscription Churn Analysis

- Trained an XGBoost churn model on 30K+ subscriber records, narrowing 60 candidate variables to 23 predictors through recursive feature elimination, mutual information scoring, and correlation pruning.
- Created decile-based churn risk bands using validation-set probabilities, PR-AUC review, and lift analysis, ranking high-risk subscribers for targeted retention planning.